SOLVER AI SDN BHD

Technical Assessment

Agentic Patient Lifecycle AI System

Extended AI Chat Capabilities for Healthcare

Issued By	Solver AI Sdn Bhd
Assessment Title	Agentic Patient Lifecycle AI System
Duration	6 Days from Date of Receipt
Submission Format	ZIP file containing all deliverables (see Section 8)
Confidentiality	This document is confidential. Do not share externally.

1. Assessment Overview

This technical assessment is issued by Solver AI Sdn Bhd as part of our engineering recruitment process. Candidates are expected to demonstrate the ability to design, prototype, and critically evaluate a production-grade Agentic Healthcare AI System.

This is not a chatbot exercise. You are expected to design an automation and control system with AI reasoning that actively manages the full patient lifecycle across five pillars:

- Pre-Visit — Intake collection, validation, and risk detection
- Post-Visit — Recovery monitoring, red-flag classification, and escalation
- Billing & Insurance — Claim integrity, duplicate prevention, and revenue protection
- Retention — Patient re-engagement, segmentation, and consent-safe outreach
- Clinical Triage — Symptom classification, urgency assessment, and safe escalation

Your submission will be evaluated on architecture depth, safety controls, workflow design, data modeling, testing rigor, and edge case handling.

2. Scenario Brief

You are building this system for a multi-branch clinic group. The following operational gaps have been identified and must be addressed:

- Incomplete intake forms causing delays and data inconsistencies
- Incorrect insurance details leading to billing rejections and revenue loss
- Missed patient follow-ups resulting in poor clinical outcomes
- Post-treatment symptoms going unreported with no automated detection
- Refills processed without contraindication checks
- Untracked billing queries resulting in lost revenue
- No structured escalation path for clinical red flags

Your system must close all of these gaps through structured agentic automation, AI-driven reasoning, and enforced escalation protocols.

3. Agent Requirements

3.1 Pre-Visit Agent

The Pre-Visit Agent is responsible for collecting and validating patient information before any clinical appointment. It must enforce data completeness and detect early risk signals.

Core Capabilities

- Collect structured intake data across all mandatory fields
- Detect and follow up on incomplete submissions automatically
- Validate insurance policy number format against defined rules
- Apply conditional logic (e.g., present pregnancy question to female patients aged 15–55)
- Score medical risk based on submitted history (e.g., hypertension + chest pain = high risk)
- Trigger nurse queue escalation for high-risk intake flags

Required Controls

- Mandatory field enforcement with structured error messages
- Insurance format validation rule (define and enforce the expected pattern)
- Risk score threshold definition and escalation trigger logic
- Automatic follow-up if submission is not completed within a defined window

3.2 Post-Visit Recovery Agent

The Post-Visit Agent proactively contacts patients after their appointment to monitor recovery and detect clinical deterioration early.

Core Capabilities

- Send automated check-in message after appointment
- Collect structured pain score (0–10) and symptom updates
- Detect red-flag phrases using NLP classification
- Support multi-turn conversation for symptom clarification
- Extract structured data from free-form patient text
- Escalate within defined SLA window upon red-flag detection

Red-Flag Classifier (Required)

You must implement a red-flag classifier capable of detecting at minimum:

- Worsening pain patterns
- Bleeding (any mention or implication)
- Shortness of breath or difficulty breathing
- Fever above defined threshold (e.g., >38.5°C)
- Medication side effects or adverse reactions

3.3 Billing & Insurance Agent

The Billing Agent protects clinic revenue by automating inquiry handling, preventing errors, and tracking outstanding claims.

Core Capabilities

- Answer patient and staff billing inquiries
- Detect potential claim rejection risk before submission
- Identify missing documentation required for valid claims
- Track and alert on unpaid invoices beyond threshold
- Prevent duplicate billing via idempotency logic

Required Controls

- Duplicate invoice detection (define unique key logic)
- Insurance policy expiry validation at time of claim
- Treatment code vs. coverage mismatch detection
- Auto-escalation for unresolved claims beyond defined SLA (e.g., >30 days)
- Manual override must require documented justification — logged to audit trail

3.4 Retention Agent

The Retention Agent re-engages inactive patients with personalised, consent-safe communication strategies.

Core Capabilities

- Identify patients with no visit in 6+ months
- Segment patients by profile: chronic, wellness, or one-time
- Trigger appropriate re-engagement message per segment
- Suppress messaging to patients with unresolved clinical issues
- Prevent outreach to patients who have not provided messaging consent

Required Controls

- Patient segmentation logic (define classification criteria)
- Suppression list system integrated with clinical and billing flags
- Consent verification before any outreach
- Communication frequency cap (e.g., max 2 contacts per month per patient)

3.5 Clinical Triage Agent

This is the highest-risk component of the system. The Clinical Triage Agent must apply strict safety guardrails at all times.

Core Capabilities

- Accept and interpret patient symptom descriptions
- Ask structured clarifying questions when information is missing or ambiguous
- Classify urgency level: Emergency, Same-Day, or Routine
- Provide safe, non-diagnostic guidance with appropriate disclaimers
- Escalate emergency keywords immediately — no delay

Strict Safety Requirements

- Must not hallucinate diagnoses or suggest definitive conclusions
- Must not give definitive medical advice — guidance only
- Must include disclaimer logic in every clinical response
- Emergency keywords (e.g., 'chest pain', 'can't breathe', 'unconscious') must trigger immediate escalation
- If information is missing → structured follow-up question required before classification

4. Architecture Requirements

4.1 System Diagram

You must submit a system architecture diagram that clearly illustrates all components and their interactions. The diagram must include:

- Chat Layer — the patient-facing interface
- Rules Engine — deterministic logic for validation, scoring, and controls
- LLM Layer — AI reasoning and NLP classification
- Data Store — schema design and storage strategy
- Escalation Engine — routing and SLA management
- Audit Log — immutable record of all system events
- Admin Dashboard — visibility and monitoring interface

4.2 Data Model (JSON Schemas)

You must define JSON schemas for all of the following entities:

- Patient
- Visit
- Intake Form
- Post-Visit Check
- Invoice
- Insurance Policy
- Clinical Alert
- Escalation Ticket

Each schema must include field names, data types, required flags, and validation rules where applicable.

5. Implementation Requirement

Your implementation must be built using n8n. Export your complete workflow as a JSON file with annotated flow logic that clearly explains each node's purpose and configuration.

Your n8n implementation must include:

- A complete exportable workflow JSON covering all five agents
- Structured input and output data at each workflow node
- Error handling branches (validation errors, upstream failures, timeouts)
- Idempotency logic — no duplicate tickets, alerts, or invoices
- Clear node naming conventions and inline annotations explaining decision logic

6. Testing Requirement

You must provide a minimum of 20 test scenarios covering all five agent pillars. Each test scenario must include:

- Input — the data or message the system receives
- Expected Output — what the system should return or do
- Escalation Status — whether escalation is triggered and to whom
- Risk Level — Low / Medium / High / Critical
- Audit Log Result — what is recorded in the audit trail

Required test distribution:

- 5 Pre-Visit scenarios
- 5 Post-Visit scenarios
- 4 Billing & Insurance scenarios
- 3 Retention scenarios
- 3 Clinical Triage scenarios

Edge cases that must be covered include but are not limited to:

- Missing or null data fields
- Conflicting data (e.g., patient confirms low pain but describes severe symptoms)
- Duplicate submissions of the same form or invoice
- User changes their answer mid-conversation

- Insurance policy found to be expired
- High pain score but patient minimises severity verbally
- Red-flag phrase embedded within a long, otherwise normal message

6.1 Reference Test Scenarios

The following table provides the minimum required test scenarios. Candidates should expand beyond these with additional edge cases.

#	Category	Scenario	Input Summary	Expected Output	Escalation	Risk Level
1	Pre-Visit	Complete valid intake	All fields filled, insurance valid	Intake accepted, confirmation sent	None	Low
2	Pre-Visit	Missing mandatory field	DOB omitted	Follow-up triggered, form flagged incomplete	None	Medium
3	Pre-Visit	Invalid insurance format	Policy no. wrong format	Validation error, patient notified to correct	None	Medium
4	Pre-Visit	High-risk medical history	Hypertension + chest pain reported	Red flag, escalate to nurse queue	Nurse Queue	High
5	Pre-Visit	Conditional field (pregnancy)	Female, age 32, pregnancy Q skipped	Agent asks pregnancy question before proceeding	None	Medium
6	Post-Visit	Normal recovery check-in	Pain score 2, no symptoms	Check-in logged, next follow-up scheduled	None	Low
7	Post-Visit	High pain but minimized	Patient says 'it's fine' but scores 8/10	System flags pain score, escalates despite dismissal	Clinical	High
8	Post-Visit	Red flag phrase in long text	Long message containing 'can't breathe properly'	NLP detects phrase, immediate escalation	Emergency	Critical
9	Post-Visit	Fever above threshold	Patient reports 39.5°C fever Day 1 post-op	Escalated to physician within SLA window	Clinical	High
10	Post-Visit	Medication side effect reported	Rash + dizziness after prescription	Alert logged, escalated to prescribing doctor	Clinical	High
11	Billing	Duplicate invoice detection	Same invoice submitted twice	Second submission blocked, audit log entry created	None	Medium
12	Billing	Expired insurance policy	Policy expired 2 months ago	Claim rejected, patient notified, billing team alerted	Billing	High
13	Billing	Treatment code mismatch	Code doesn't match coverage plan	Mismatch flagged, claim held pending review	Billing	High
14	Billing	Unresolved claim >30 days	Claim open for 35 days	Auto-escalation to billing manager, SLA breach logged	Billing Mgr	High
15	Retention	Inactive patient 6+ months	No visit in 8 months, chronic patient	Re-engagement message sent per segment	None	Low

16	Retention	Patient on suppression list	Flagged with unresolved clinical issue	Messaging suppressed, no outreach until resolved	None	Medium
17	Retention	Frequency cap exceeded	Patient already contacted 3x this month	Outreach blocked, next scheduled next cycle	None	Low
18	Clinical	Emergency keyword detected	Patient says 'I think I'm having a heart attack'	Immediate emergency escalation, call 999 prompt	Emergency	Critical
19	Clinical	Routine symptom inquiry	Mild headache for 2 days, no red flags	Classified as routine, book next available slot	None	Low
20	Clinical	Ambiguous symptom, missing info	Patient says 'chest feels weird'	Agent asks structured follow-up questions before classifying	Pending	Medium

7. Reporting & Dashboard Design

Design a concept dashboard that provides operational visibility across all five agents. The dashboard must surface the following metrics at minimum:

- Red flag cases — volume, category, and resolution status
- Unresolved billing claims — age, value, and escalation status
- High-risk patients — flagged intake, active clinical alerts
- SLA breaches — by agent type and time overdue
- Retention conversion rate — outreach sent vs. re-appointments booked
- Escalation volume — broken down by agent and escalation type

Submit your dashboard design as a wireframe, annotated mockup, or component specification.

8. Submission Requirements

Submit a single ZIP file named: LastName_FirstName_SolverAI_Assessment.zip

The ZIP must contain:

- README.md — setup instructions, technology decisions, and known limitations
- Architecture Diagram — PDF, PNG, or editable format (Lucidchart, Draw.io, Figma, etc.)
- JSON Schemas — all 8 data models as defined in Section 4.2
- Implementation — n8n workflow exported as JSON with annotated flow logic
- Test Scenarios — all 20+ scenarios in tabular or structured format (CSV, Markdown, or JSON)
- Dashboard Design — wireframe or annotated mockup

- Video Presentation — a screen-recorded walkthrough (with voice-over) demonstrating your implemented workflows, live testing of all five agents, and key design decisions. Maximum 15 minutes. Accepted formats: MP4 or MOV.
- Bonus Features (if applicable) — clearly labelled in README

You have 6 days from the date of receipt of this document to complete and submit your assessment. Late submissions will not be reviewed unless an extension has been granted in writing by Solver AI Sdn Bhd.

9. Terms & Confidentiality

- This assessment is confidential and intended solely for the named recipient.
- Do not share this document or your submission with third parties.
- All work submitted must be your own. Use of AI assistance is permitted for implementation but must be disclosed in your README.
- Solver AI Sdn Bhd reserves the right to discuss your submission in a follow-up technical interview.
- Submitted work will not be used for commercial purposes.